A unified and simplified treatment of the non-linear equilibrium problem of double-row rolling bearings. Part 1: rolling bearing model

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Abstract: To increase the rigidity of bearings–shaft systems, the shaft is often supported by a series of double-row rolling bearings. Little work on multiple-row rolling bearings has been published. A comprehensive model is proposed which permits the determination of the internal interactions. The total elastic deflection between bearing rings is described using a vector-and-matrix method. A variety of double-row rolling bearing types are analysed, such as tapered roller bearing, cylindrical roller bearing, spherical roller bearing, self-aligning ball bearings and angular contact ball bearing. The basic internal geometry (including the internal clearance) and the effect of the initial preload (in term of the initial axial compression) are considered.

Keywords: double-row rolling bearings, five degrees of freedom, load distribution, vector-and-matrix method

NOTATION

A	distance between the centre of curvature of the inner and outer raceways (m)
ACBB	angular contact ball bearing
CRB	cylindrical roller bearing
d_{m}	pitch circle diameter (m)
D	rolling element diameter (m)
DB	back-to-back arrangement
DF	face-to-face arrangement
e	axial distance between the roller inertial frame origins of the bearing rows (m)
E'	effective elastic modulus (N/m^2)
F_x, F_y, F_z	bearing loads applied along the x, y and z directions respectively (N)
K	load–deflection constant for a point contact (N/m$^{3/2}$) and for a line contact (defined per unit length) (N/m$^{19/9}$)
L	roller effective length (m)
$\boldsymbol{M}$	bearing tilting moment vector
n	exponent depending on the contact type = 3/2 for point contact and = 10/9 for line contact
$\boldsymbol{n}$	unit vector
q	contact load intensity (N/m)
$\boldsymbol{q}$	contact load intensity vector
Q	contact load (N)
$\boldsymbol{Q}$	contact load vector
r_{i}	inner raceway curvature radius (m)
$\boldsymbol{r}$	nominal position vector in the inertial frame
$\boldsymbol{r}'$	displaced position vector in the inertial frame
R	distance from the rolling element origin frame to bearing axis (m)
SABB	self-aligning ball bearing
SRB	spherical roller bearing
[**T**]	transformation matrix from roller to inertial frame
TRB	tapered roller bearing
u	clearance (m)
x	roller axial coordinate (m)
(x, y, z)	rolling element frame
(X, Y, Z)	bearing inertial frame
α	contact angle (rad)

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δ_p	axial clearance/compression of the bearing row (m)
$\delta_x, \delta_y, \delta_z$	relative linear displacements of the bearing along the x, y and z directions respectively (m)
$\boldsymbol{\delta}$	bearing displacement vector
Δ	total approach between bearing rings (m)
ε	half tapered angle of roller (rad)
θ_y, θ_z	relative tilting angles of bearing around the y and z directions respectively (rad)
$[\boldsymbol{\theta}]$	outer displaced to outer fixed transformation matrix
$\boldsymbol{\rho}$	position vector for outer raceway in roller frame
$\sum \rho$	curvature sum (m^{-1})
ψ	angular position of a rolling element (rad)

Subscripts

a	axial
i	inner ring
m	bearing row
o	outer ring
r	radial
s	component of the bearing load vector

1 INTRODUCTION

In some applications where double-row rolling bearings are used such as gear boxes, machine tool spindles or rotor systems, knowledge of the load distribution among the rolling elements as well as the variation along the length of the individual rolling element is of prime importance. This involves taking into consideration the non-linearity of both the contact load–deformation of the rolling element–raceway and the bearing equilibrium equations.

The rolling element equilibrium has become a subject of investigation performed by many researchers and designers. The rolling element literature gives three methods to determine the internal load distribution:

Method 1. The first way consists of a global treatment of the bearing when the load distribution can be analysed in terms of the total approach between rings. Based on this methodology, theoretical research on mechanical interactions in rolling bearings (only radially loaded) can be traced back to 1907 by Stribeck. Later, Harris **[1]** analysed the static equilibrium for rolling bearings loaded with two degrees of freedom, using the concept of Sjovall's integrals **[2]**. Recently, Houpert **[3]** has extended Harris' work for single-row ball and roller bearings loaded with five degrees of freedom where input data are the relative displacements between rings and not forces and moments.

Method 2. In a second way, both the individual interactions and the stiffness of each Hertzian rolling element–raceways contact are taken into account. The position of each rolling element within the bearing is linked to three unknowns per roller (two translations and a rotation) for roller bearings and two unknowns per ball (two translations) for ball bearings. In other words, this method requires us to introduce $3 \times Z$ supplementary unknowns for roller bearings and $2 \times Z$ supplementary unknowns for ball bearings, Z being the number of rolling elements. Considerable work has been devoted to the development of analytical procedures based on this method. Liu **[4]** proposed a nice analytical derivation for the equilibrium analysis of tapered roller bearings considering five relative displacements for the inner ring. De Mul *et al.* **[5, 6]** modernized and extended Liu's model for ball and roller bearings. These workers defined the Jacobian matrices for both the rolling elements and the bearing. More recently, Bourdon *et al.* **[7, 8]** presented a general methodology for bearing static non-linear behaviour. The method consists in defining one 'element' (in the meaning of the finite element analysis) for each rolling element of the bearing considered. Generally, the models developed using this methodology are quite sophisticated and require three equilibrium equations for each ring and also three for each rolling element.

Method 3. The third and most powerful way, but almost the most sophisticated requiring heavy computing and data analysis with complex input data, consists in solving simultaneously the six dynamic equilibrium equations for each bearing element, including the cage, as done by Gupta **[9]**. This approach is more adapted for insight analysis of a single rolling element bearing, independently of its environment.

Starting from these previous studies, this paper proposes a new general methodology for modelling the static mechanical interactions in all types of double-row rolling bearings. The model presents a combination between methods 1 and 2 presented above. Thus, the theoretical formulation can be divided in two parts:

1. Firstly, using a vector-and-matrix method, the total geometrical interactions (total approach between rings) will be formulated.
2. Secondly, the normal contact force will be computed using only the equilibrium of one of the bearing rings (i.e. the inner or the outer ring).

The conventional Hertzian elastic contact theory is considered to express the relationship between contact load and deflection. For a roller bearing it should be mentioned that the present analysis is valid for roller and raceway profiles rectilinear (without correction) or more complex (crowned, logarithmic, etc.) since the slicing technique is used to determine both the contact pressure distribution and the related elastic deflection. A simple rectilinear profile is sufficient to forecast the shaft deflection but might lead to an underestimation of maximum stress levels near the worst loaded roller ends under moment loads, as previously shown in the literature, for example by Ioannou *et al.* [**10**].

Using this method, the computing time remains limited, the explicit equilibrium of each rolling element being removed.

2 GEOMETRICAL PARAMETERS

Types of double-row rolling bearings that will be considered are:

(a) tapered roller bearings (TRBs),
(b) cylindrical roller bearings (CRBs),
(c) spherical roller bearings (SRBs),
(d) self-aligning ball bearings (SABBs) and
(e) angular contact ball bearings (ACBBs).

The main geometrical parameters used in the present analysis include the rolling element diameter D and in the case of roller bearings the roller effective length L, the number of rolling elements, Z, the pitch circle diameter d_{m}, the raceway groove curvature radii r_{i} and r_{o}, the contact angle α and the radial clearance u.

3 TOTAL RELATIVE APPROACH BETWEEN RINGS

The vector-and-matrix description is used to describe how displacements are distributed between rolling elements. For convenience the following coordinate systems are introduced:

(a) an inertial coordinate system (X, Y, Z) with respect to the inner ring axis and with its origin fixed to the initial bearing centre;
(b) two local coordinate systems (x_1, y_1, z_1) and (x_2, y_2, z_2), which are attached to the geometrical centre of each of the rolling elements of rows 1 and 2 respectively, except for each of the taper rollers for which the system origin corresponds to the centre of the roller middle section.

All coordinate systems are right handed. Usually, it is considered that rolling bearings are externally loaded by the inner ring and the bearing equilibrium is conveniently analysed by keeping its outer ring fixed in space. For the general case, the bearing is loaded in five degrees of freedom: three forces F_x, F_y and F_z, and two tilting moments M_y and M_z. Consequently, the inner ring moves through linear displacements δ_x, δ_y and δ_z and angular rotations θ_y and θ_z. Mechanical interactions between bearing elements are basically the same when the inner ring is assumed fixed in space and the outer ring displaced of $(-\delta_x, -\delta_y, -\delta_z, -\theta_y, -\theta_z)$ with respect to the X, Y and Z axes. To simplify the analysis, the second situation will be considered for any type of rolling bearing mentioned above.

Firstly, the vector and matrix relations to determine the total approach between rings will be presented for the main types of double-row rolling bearings, as shown in Fig. 1. The locations of the outer raceway points contact are, for convenience, described using position vectors indicated with generic symbols $\boldsymbol{r}_{\mathrm{o}}$ for the nominal positions and with $\boldsymbol{r}'_{\mathrm{o}}$ for the displaced positions, the vectors being expressed in the (X, Y, Z) inertial system. The total approach (deformation) between bearing rings can now be expressed as

$$\Delta_m = \left(\boldsymbol{r}'_{\mathrm{o}m} - \boldsymbol{r}_{\mathrm{o}m}\right) \cdot \boldsymbol{n}_{\mathrm{o}m} \tag{1}$$

where the subscripts o and m refer to the outer raceway and to the bearing row ($m=1$ for the left-hand row and $m=2$ for the right-hand row) and $\boldsymbol{n}_{\mathrm{o}m}$ represents the unit vector perpendicular to the surface, as shown in Fig. 2 for the interaction between the roller and the outer ring.

For a contact point on the outer raceway, the final position vector $\boldsymbol{r}'_{\mathrm{o}}$ is expressed in the (X, Y, Z) inertial frame as

$$\boldsymbol{r}'_{\mathrm{o}m} = \boldsymbol{r}_{\mathrm{o}m} + \boldsymbol{\delta}_{\mathrm{o}} + [\boldsymbol{\theta}_{\mathrm{o}}]\boldsymbol{r}_{\mathrm{o}m} \tag{2}$$

in which $\boldsymbol{\delta}_{\mathrm{o}}$ represents the displacement vector of the outer ring and $[\boldsymbol{\theta}_{\mathrm{o}}]$ represents the outer rotation transformation matrix.

Then, for the same contact point, the nominal position vector $\boldsymbol{r}_{\mathrm{o}}$ can be expressed in the (X, Y, Z) inertial system by

$$\boldsymbol{r}_{\mathrm{o},m} = \boldsymbol{r}_{\mathrm{b}m} + [\mathbf{T}_m]\boldsymbol{\rho}_{\mathrm{o}m} \tag{3}$$

where $\boldsymbol{r}_{\mathrm{b}m}$ is, in the inertial system, the position vector of the roller system origin; the vector $\boldsymbol{\rho}_{\mathrm{o}m}$ is located in the roller system (x_m, y_m, z_m), with $m=1$ or 2, the points coming into contact from the outer raceway. $[\mathbf{T}_m]$ is the transformation matrix from the (x_m, y_m, z_m) to the (X, Y, Z) coordinate system. By combining equations (1) to (3), the total relative approach (or deformation) Δ_m can now be obtained as

$$\Delta_m = [\boldsymbol{\delta}_{\mathrm{o}} + [\boldsymbol{\theta}_{\mathrm{o}}](\boldsymbol{r}_{\mathrm{b}m} + [\mathbf{T}_m]\boldsymbol{\rho}_{\mathrm{o}m})] \cdot \boldsymbol{n}_{\mathrm{o}m} \tag{4}$$

According to each rolling element type represented in Fig. 1, the vectors $\boldsymbol{\rho}_{\mathrm{o}m}, \boldsymbol{r}_{\mathrm{b}m}$ and $\boldsymbol{n}_{\mathrm{o}m}$ and the transformation matrices $[\boldsymbol{\theta}_{\mathrm{o}}]$ and $[\mathbf{T}_m]$ are given as follows.

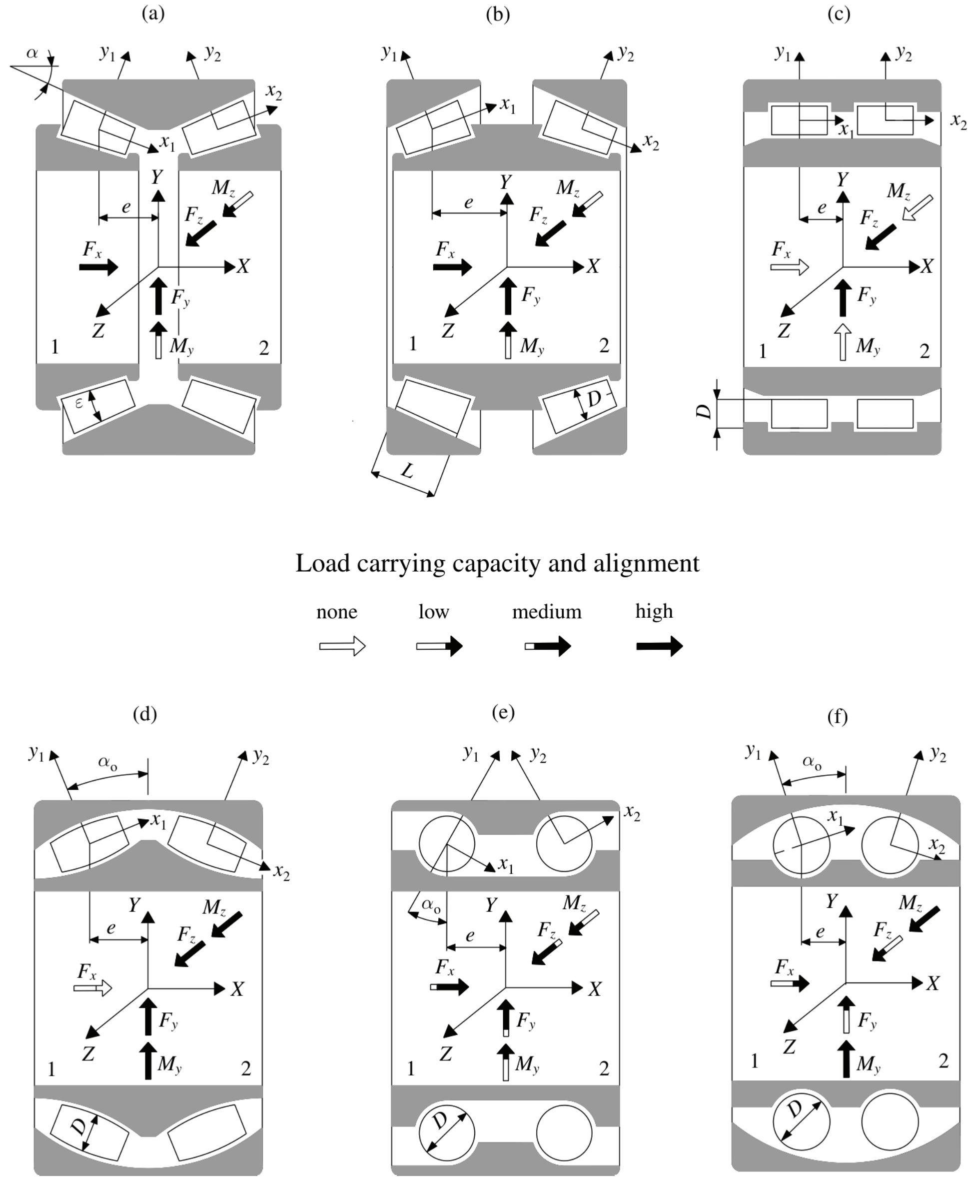

Fig. 1 Selection of double-row rolling bearings subjected to analysis and corresponding coordinate systems: (a) TRB (DB); (b) TRB (DF); (c) CRB; (d) SRB; (e) SABB; (f) ACBB

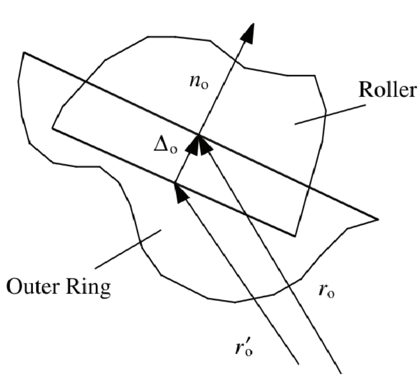

Fig. 2 Roller–outer ring interaction

3.1 Definition of vectors

$$\delta_o = -\begin{bmatrix} c_1(\delta_x + c_2\delta_p) \\ \delta_y \\ \delta_z \end{bmatrix} \tag{5}$$

where δ_p represents the initial axial compression between bearing rows; the coefficients c_1 and c_2 are dependent on bearing type as

$c_1 = 1$ for TRBs, ACBBs and SRBs
$c_1 = 0$ for CRBs

$$c_2 = 0 \text{ for CRBs, ACBBs and SRBs}$$
$$c_2 = \pm 1 \text{ for TRBs (DB)}$$
$$c_2 = \mp 1 \text{ for TRBs (DF)}$$

(In the last two the upper sign is for the left-hand row and the lower sign for the right-hand row.)

$$\boldsymbol{r}_{bm} = \begin{bmatrix} c_3 e \\ R\cos\psi \\ R\sin\psi \end{bmatrix} \tag{6}$$

R is the distance from the origin of the rolling element system to the bearing axis, ψ is the rolling element position angle and the coefficient c_3 is correlated with the index m. So, $c_3 = -1$ for the left-hand row (when $m = 1$), and $c_3 = 1$ for the right-hand row (when $m = 2$).

$$\boldsymbol{\rho}_{om} = \begin{bmatrix} -c_3 x \\ \dfrac{D}{2} - c_4 x \tan\varepsilon \\ 0 \end{bmatrix} \tag{7}$$

D represents the roller mean diameter for TRBs, the roller maximum diameter for SRBs and the rolling element diameter for CRBs, ACBBs and SABBs. x is the length coordinate along the roller axis and ε is the half tapered angle of roller for TRBs (in the case of the other bearings, $\varepsilon = 0$) (see Fig. 1).

$$\boldsymbol{n}_{om} = \begin{bmatrix} c_3 c_4 \sin\alpha \\ -\cos\alpha\cos\psi \\ -\cos\alpha\sin\psi \end{bmatrix} \tag{8}$$

The coefficient c_4 depends on the arrangement of rows; thus, $c_4 = 1$ for ACBBs and TRBs (DB), and $c_4 = -1$ for SRBs, SABBs and TRBs (DF). In the case of CRBs the contact angle α is zero and, thus, equation (8) does not depend on the coefficients c_3 and c_4.

3.2 Definition of matrices

Using the small-angle approximations, the transformation matrix $[\boldsymbol{\theta}_o]$ due to misalignment is defined as

$$[\boldsymbol{\theta}_o] = \begin{bmatrix} 0 & \theta_z & -\theta_y \\ -\theta_z & 0 & 0 \\ \theta_y & 0 & 0 \end{bmatrix} \tag{9}$$

The transformation matrix $[\mathbf{T}_m]$ from the rolling element system of each row to the inertial system is defined by

$$[\mathbf{T}_m] = \begin{bmatrix} \cos\alpha & -c_3 c_4 \sin\alpha & 0 \\ c_3 c_4 \sin\alpha\cos\psi & \cos\alpha\cos\psi & -\sin\psi \\ c_3 c_4 \sin\alpha\sin\psi & \cos\alpha\sin\psi & \cos\psi \end{bmatrix} \tag{10}$$

Substituting equations (5) to (10) into equation (4) yields the following general expression for the total approach between raceways:

$$\begin{aligned} \Delta_m = &- c_1 c_3 c_4 (\delta_x + c_2 \delta_p)\sin\alpha \\ &+ (\delta_y + c_3 a\theta_z)\cos\alpha\cos\psi \\ &+ (\delta_z - c_3 a\theta_y)\cos\alpha\sin\psi \\ &+ x c_3 b(\theta_y \sin\psi - \theta_z\cos\psi) \end{aligned} \tag{11}$$

where a and b are geometrical parameters depending on the bearing construction type:

$$a = c_4\left(R\tan\alpha + \frac{D}{2}\frac{\sin\varepsilon}{\cos\alpha}\right) + e \tag{12}$$

Particularizing equation (11) for different types of bearing gives

$$a = \begin{cases} e + Rm\tan\alpha + \dfrac{D}{2}\dfrac{\sin\varepsilon}{\cos\alpha} & \text{for TRBs (BD)} \\ e - \left(Rm\tan\alpha + \dfrac{D}{2}\dfrac{\sin\varepsilon}{\cos\alpha}\right) & \text{for TRBs (BF)} \\ Rm\tan\alpha + e & \text{for ACBBs} \\ e & \text{for CRBs} \\ 0 & \text{for SRBs and SABBs} \end{cases} \tag{13}$$

and

$$b = \begin{cases} \dfrac{1}{\cos\varepsilon} & \text{for TRBs} \\ 1 & \text{for CRBs} \end{cases} \tag{14}$$

Denoting by $\delta_x^*, \delta_y^*, \delta_z^*$ the resulting approach, and by θ^* the total effect of misalignments θ_y and θ_z gives

$$\delta_{xm}^* = -c_1 c_3 c_4(\delta_x + c_2\delta_p) \tag{15a}$$
$$\delta_{ym}^* = (\delta_y + c_3 a\theta_z) \tag{15b}$$
$$\delta_{zm}^* = (\delta_z - c_3 a\theta_y) \tag{15c}$$
$$\theta_m^* = c_3 b\big(\theta_y\sin\psi - \theta_z\cos\psi\big) \tag{15d}$$

Equation (11) can be now rewritten as

$$\Delta_m = \Delta_m^o + x\theta_m^* \tag{16}$$

and for the total approach between raceways in the middle plane ($x = 0$)

$$\begin{aligned} \Delta_m^o = &\delta_{xm}^*\sin\alpha + \delta_{ym}^*\cos\alpha\cos\psi \\ &+ \delta_{zm}^*\cos\alpha\sin\psi \end{aligned} \tag{17}$$

3.3 Initial clearance

Generally, double-row rolling bearings are assembled with a certain amount of diametral clearance. To consider the effect of the initial clearance on the total approach, displacement terms found in equations (11) to (17) must be adapted to each bearing construction type.

Except for CRBs, the radial clearance u_r is linked to the axial clearance u_a. Thus, in the case of TRBs, ACBBs, SRBs and SABBs, localized contact elastic deflections will occur only when the axial clearance vanishes. For CRBs, the contact elastic deflection begins when the radial clearance is removed. As function of bearing type, the internal clearance affects equation (11) as follows:

clearance for TRBs, ACBBs, SRBs and SABBs is

$$\Delta_m^o = (\delta_{xm}^* - u_a)\sin\alpha + \delta_{ym}^*\cos\alpha\cos\psi + \delta_{zm}^*\cos\alpha\sin\psi \tag{18}$$

and for CRBs is

$$\Delta_m^o = \delta_{xm}^* + \delta_{ym}^*\cos\psi + \delta_{zm}^*\sin\psi - u_r \tag{19}$$

For CRBs, the contact angle is nil and the clearance affects only the radial components of the total approach between raceways.

4 STATIC LOADS IN CONCENTRATED CONTACTS

When the total approach between rings Δ has been determined, the contact loads Q can be evaluated for dry contacts by the Hertzian theory for elastic deformation:

$$Q_j = \max[0, K(\Delta_j^o)^n] \tag{20}$$

where n is an exponent depending on the contact type [**11**] ($n = 10/9$ for a line contact [**12**] and $n = 3/2$ for a point contact) and K is the total stiffness constant given by Harris [**11**]:

$$K = \left[\left(\frac{1}{K_i}\right)^{1/n} + \left(\frac{1}{K_o}\right)^{1/n}\right]^{-n} \tag{21}$$

in which K_i and K_o represent the stiffness constants of the inner and the outer roller contacts respectively. According to SI units (newtons and metres), the stiffness constant for bearing line contact can be estimated as described by Palmgren [**12**]:

$$K_i = K_o = 0.356E'L^{8/9} \tag{22}$$

where L is the contact length and E' the effective elastic modulus.

In the case of a point contact, the stiffness coefficient given by Harris [**11**] is

$$K_{i,o} = \frac{2\sqrt{2}}{3}\frac{E'}{\delta_{i,o}^* \sum\rho_{i,o}^{1/2}} \tag{23}$$

where $\Sigma\rho$ is the curvature sum at the contact point and δ^* is the dimensionless contact deformation given by Harris [**11**] as a function of the curvature function $F(\rho)$. In the case of ACBBs or roller bearings, K from equation (23) varies with contact angle as discussed by Harris [**11**].

5 BEARING EQUILIBRIUM EQUATIONS

Once the contact forces are defined, the static equilibrium equations for the outer ring may be expressed in the (X, Y, Z) inertial coordinate system. For the general case of bearings loaded with five degrees of freedom, the external forces and moments are denoted by $F_x = F_1$, $F_y = F_2$, $F_z = F_3$, $M_y = F_4$ and $M_z = F_5$. The balance equations will be written for bearings containing both point and line contacts.

5.1 Bearings with point contacts

Using a vector base $\boldsymbol{x}_1$, $\boldsymbol{x}_2$, $\boldsymbol{x}_3$, the contact force $\boldsymbol{Q}_m$ applied by a roller acting on the outer ring becomes

$$\boldsymbol{Q}_m = Q_m\boldsymbol{n}_m = Q_m\begin{bmatrix} c_3c_4\sin\alpha \\ -\cos\alpha\cos\psi \\ -\cos\alpha\sin\psi \end{bmatrix} \tag{24}$$

and the tilting moment around the origin of the (X, Y, Z) inertial system is

$$\boldsymbol{M}_m = Q_m\boldsymbol{r}_{om} = -Q_m\begin{bmatrix} 0 \\ c_4a\cos\alpha\sin\psi \\ -c_4a\cos\alpha\cos\psi \end{bmatrix} \tag{25}$$

Therefore the equilibrium equations of the outer ring are

$$\boldsymbol{F} - \sum_{m=1}^{2}\frac{Z}{2\pi}\int_0^{2\pi}\boldsymbol{G}\times\boldsymbol{Q}\,\mathrm{d}\psi = \boldsymbol{0} \tag{26}$$

where $\boldsymbol{F}$ is the force vector given by

$$\boldsymbol{F} = [\boldsymbol{F}_s], \quad s = \overline{1,5}$$

and

$$\boldsymbol{G} = [\boldsymbol{G}_{ms}] = \begin{bmatrix} c_3c_4\sin\alpha \\ -\cos\alpha\cos\psi \\ -\cos\alpha\sin\psi \\ t\sin\psi \\ -t\cos\psi \end{bmatrix}$$

with $t = c_4a\cos\alpha$ and $\boldsymbol{Q} = [Q_m]$, $m = \overline{1,2}$.

The balance equations can be easily adapted [cf. equation (28)] to different bearing types having point contacts (ACBBs, SRBs or SABBs) by choosing adequate values of the coefficients c_3 and c_4. The contact operating angle α at any rolling element position is given in the Appendix.

5.2 Bearings with line contacts

For bearings with line contacts (TRBs or CRBs) the contact load vector $\boldsymbol{Q}_m$ has the same components as defined for bearings with point contact, but the exponent n becomes 10/9. In the case of TRBs the tilting moment around the origin of the (X, Y, Z) inertial system can be defined as

$$\boldsymbol{M} = \int_0^{2\pi} \int_{-L/2}^{L/2} (\boldsymbol{q}_m \times \boldsymbol{r}_{om})\, \mathrm{d}x\, \mathrm{d}\psi \qquad (27)$$

in which $\boldsymbol{q}$ represents the vector of the contact load intensity given by

$$\boldsymbol{q}_m = q_m \boldsymbol{n}_m = -q_m \begin{bmatrix} c_3 c_4 \sin\alpha \\ -\cos\alpha\cos\psi \\ -\cos\alpha\sin\psi \end{bmatrix} \qquad (28)$$

Substituting equation (28) into equation (27) gives

$$\boldsymbol{M}_m = -\frac{Z}{2\pi} \int_0^{2\pi} \int_{-L/2}^{L/2} q_m \begin{bmatrix} 0 \\ t(x)\sin\psi \\ -t(x)\cos\psi \end{bmatrix} \mathrm{d}\psi\, \mathrm{d}x \qquad (29)$$

where

$$t(x) = c_4 a \cos\alpha - xb$$

The balance equations of the outer ring can now be written as

$$\boldsymbol{F} - \sum_{m=1}^{2} \frac{Z}{2\pi} \int_0^{2\pi} \int_{-L/2}^{L/2} \boldsymbol{G} \times \boldsymbol{q}\, \mathrm{d}\psi = \boldsymbol{0} \qquad (30)$$

where $\boldsymbol{F}$ is the force vector given by

$$\boldsymbol{F} = [\boldsymbol{F}_s], \qquad s = \overline{1,5}$$

$$\boldsymbol{G} = [\boldsymbol{G}_{ms}] = \begin{bmatrix} c_3 c_4 \sin\alpha \\ -\cos\alpha\cos\psi \\ -\cos\alpha\sin\psi \\ t(x)\sin\psi \\ -t(x)\cos\psi \end{bmatrix}$$

$$\boldsymbol{q} = [q_m], \qquad m = \overline{1,2}$$

The load intensity q_m is given by

$$q_m = \frac{K}{L}(\varDelta)^n = \frac{K}{L}(\varDelta_m^{\circ} + x\theta_m^*)^n \qquad (31)$$

Applying a first-order perturbation, the contact load intensity can be expressed as

$$q_m \approx \frac{K}{L}\left[(\varDelta_m^{\circ})^n + xn(\varDelta)^{n-1}\theta_m^*\right]$$

Combining equations (29) and (31) and integrating with respect to the x coordinate gives

$$\boldsymbol{M} = \sum_{m=1}^{2} \frac{Z}{2\pi} \int_0^{2\pi} \begin{bmatrix} 0 \\ g\sin\alpha \\ -g\sin\alpha \end{bmatrix} \mathrm{d}\psi \qquad (32)$$

with

$$g = K\left(-\frac{nbL^2(\varDelta_m)^{n-1}\theta_m^*}{12} + c_4 a(\varDelta_m)^n \cos\alpha\right)$$

The systems of equations (26) and (30) for bearings having point and line contact respectively are non-linear and can be solved by the Newton–Raphson iterative method.

6 CONCLUSIONS

A generalized formulation for the static non-linear behaviour of several types of double-row rolling bearing (TRBs, CRBs, SRBs, SABBs and ACBBs) is presented. The generality of the method is such that it can be readily adapted to all types of single-row rolling bearing. An analytical framework for the computation of applied forces and moments at the rolling element–race interaction has been described.

Bearing equilibrium is obtained only by solving the equilibrium of one ring (such as the outer ring). Thus, with this model, the individual equilibrium of each rolling element is removed and, as a result, computing time reduced. Part 2 of this two-part paper will present an application example of a flexible shaft with straddle and cantilever mounting with double-row tapered roller bearings (DRTRBs in both DB and DF arrangements).

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APPENDIX

Except for bearings with line contacts (TRBs and CRBs) the operating contact angle α at any rolling element at position ψ can be determined from

$$\alpha_m = a\tan\left(\frac{\delta^*_{xm} + R_\mathrm{i}\theta^*_m + A\sin\alpha_\mathrm{o}}{\delta^*_{ym}\cos\psi + \delta^*_{zm}\sin\psi + A\cos\alpha_\mathrm{o} - r_\mathrm{i}\theta^*_m\sin\alpha}\right) \quad (33)$$

where r_i is the inner raceway groove curvature radius, R_i is the locus of the inner ring raceway groove curvature radius as defined by

$$R_\mathrm{i} = 0.5d_m + (r_\mathrm{i} - 0.5D)\cos\alpha_\mathrm{o} \quad (34)$$

and m is the index referring to bearing row ($m = 1$ for the left-hand row and $m = 2$ for the right-hand row). A represents the distance between the centre of curvature of the inner and outer rings:

$$A = R_\mathrm{i} + R_\mathrm{o} - D \quad (35)$$